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Analyzing Disciplinary Variation in Abstracts: Linguistic Features across Scientific and Technical Fields

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Abstract This study investigated the linguistic features of abstracts across five scientific and technical disciplines: biochemistry, civil engineering, computer and information sciences, electronics engineering, and mechanical engineering. Abstracts play a crucial role in academic papers by summarizing key findings, methodologies, and implications. Through the lens of English for Specific Purposes (ESP) and genre analysis, this study aimed to determine whether linguistic features vary across disciplines and classify these disciplines based on the similarity of their linguistic features. The corpus consisted of 5,000 abstracts, with 1,000 from each discipline, sourced from the open-access journal PLOS ONE. Using Biber's [1] multidimensional analysis framework, this study examined 63 of 67 linguistic features, including passive voice, conjunctions, amplifiers, and discourse markers. Statistical analysis, including correlation and cluster analyses, revealed that the disciplines can be broadly divided into two groups: biochemistry and computer & information sciences, and a second group including mechanical engineering, civil engineering, and electronics engineering. These findings suggest that while some linguistic features are shared across disciplines, others vary substantially. For example, biochemistry has a higher frequency of passive constructions and large noun phrases, whereas computer and information sciences frequently use first-person pronouns and amplifiers. These insights are valuable for ESP instruction as they highlight the need for discipline-specific writing guidance in higher education. Educators can use this information to develop effective writing instruction tailored to the linguistic norms of each field. Future research could expand on these findings by exploring additional rhetorical elements and examining the impact of linguistic features on reader comprehension.

Keywords English for Specific Purposes, English for Science and Engineering, Corpus Analysis, Disciplinary Variation Analysis

1. Introduction

English has long been recognized as a *lingua franca* in scientific and technical fields [2-5]. Studies have shown that over 90% of papers published in this field are written in English [6], and this proportion has steadily increased over the past few decades [7]. Consequently, researchers from non-English-speaking countries mainly publish their work in English, underscoring the dominant role of language in global scientific communication.

The linguistic characteristics of the scientific and technical fields have attracted significant attention in the context of ESP. ESP research is fundamentally aimed at providing language instruction tailored to the learners' specific needs and goals. In scientific and technical fields, these tailored approaches have added significance because

of the specialized language and communication practices required. Findings from ESP studies have been widely implemented in higher education institutions to enhance students' ability to navigate the linguistic demands of their chosen fields [8].

In educational settings, significant attention has been paid to understanding how texts are structured and organized within particular contexts. This approach, known as genre analysis, focuses on identifying and analyzing the distinctive features of different types of texts (genres) to understand how they function and convey meaning within specific communities or disciplines. Genre analysis not only helps learners understand the norms and structures of various academic texts (e.g., papers, reports, essays) but also aids in developing targeted ESP curricula and prepares students for effective academic communication.

Genre analysis is crucial in scientific and technical fields because these disciplines demand highly specialized ways of presenting information. Clarity, precision, and adherence to specific norms are essential for effective communication and dissemination of research findings. Identifying these disciplinary conventions can enhance the understanding and production of academic texts within these disciplines. Understanding the characteristics of texts in a target discipline allows learners to anticipate their structure and focus on the key elements, thereby making it easier to extract relevant information and understand the author's intentions.

Several studies have shown that academic fields possess distinct genres and norms [9,10]. Structures of scientific and technical texts are expected to vary across disciplines. Noguchi & Kunioshi [11] emphasize the role of genre-based, corpus-supported writing courses for science and engineering students in Japan, demonstrating how corpus use helps learners master writing conventions specific to their disciplines. This underscores the importance of genre-based instruction in shaping students' academic writing skills and meeting the demands of scientific discourse. Comparing linguistic features across disciplines is essential to identify shared and discipline-specific communication strategies. Such a disciplinary variation analysis reveals whether certain linguistic features are universal in scientific writing or shaped by the nature of the discipline, providing insights that can inform tailored writing instructions for each field.

Abstracts are one of the most important components of academic papers and provide a concise summary of the research. They serve as the primary gateway to a research paper, concisely distilling the essence of the study. Their critical role in conveying key findings, methodologies, and implications makes them valuable sites for examining linguistic features reflecting both universal scientific discourses and discipline-specific norms. Since abstracts are often the first sections read by other researchers, the linguistic features used —such as vocabulary, grammar, and discourses structures— are critical for ensuring clear and accurate communication of research findings. Therefore, it is important to understand how these features vary across scientific and technical disciplines.

Specifically, to successfully integrate into an academic community, as shown by Suzina [12], genre analysis enables students to master vocabulary, grammar, syntax, and discourse patterns critical to meeting community standards.

Traditional ESP research has tended to focus on vocabulary and phrases. For example, Coxhead [13] developed the Academic Word List, which includes 570 word families commonly found in academic texts, but not covered by the General Service List, which was developed from a wide variety of written and spoken contexts. Her work involved analyzing a 3.5-million-word corpus across various academic disciplines to identify words frequently used across fields, providing a valuable resource for students aiming to enhance their academic vocabulary. Similarly, Lam [14] studied how second language learners acquire English vocabulary through reading and writing. His research focused on the effectiveness of integrating these tasks to facilitate vocabulary acquisition, demonstrating the importance of active engagement with language for vocabulary development.

In the specific context of science and technology, Farrell [15] performed a lexical analysis of texts in the field of electronics, identifying technical terms unique to the field and semi-technical vocabulary that, although not exclusive to electronics, remain crucial for understanding the texts within this area.

Graham [16] introduced the Engineering Phrases List; it was developed by analyzing engineering textbooks, research papers, and other relevant materials. It emphasizes the importance of phrases over individual words, recognizing that certain phrases are recurrent and essential in engineering communication.

Heidari et al. [17] highlighted the importance of understanding both vocabulary and rhetorical strategies unique to each academic field, showing that disciplines prioritize different moves (e.g., establishing a niche, reviewing literature) in research article introductions.

Although vocabulary and phraseology are vital for language acquisition, it is equally important to focus on grammatical structures and discourse patterns. These elements shape the flow and logic of a text, particularly in scientific writing.

As Biber et al. [18] emphasize, the intersection of grammar and discourse plays a crucial role in communication across genres. Hyland [19] builds on this by emphasizing the importance of metadiscourse, which guides readers through complex text. He suggests that understanding these elements is crucial to achieve coherence and clarity in writing.

Given the importance of understanding the linguistic features specific to each discipline, the following research questions guided this study:

RQ1. What are the linguistic variations in abstracts across different scientific disciplines?

RQ2. Is it possible to classify abstracts in scientific and technical fields based on the similarity of their linguistic features?

By addressing the research questions posed, this study aims to provide actionable insights for educators and curriculum developers in science and technology. Understanding the linguistic features that differ or remain consistent across disciplines can inform more effective teaching practices, particularly in ESP contexts where the ability to navigate the complexities of scientific communication is important.

2. Data and Methods

2.1. Data

The titles and abstracts of scientific and technological papers were collected from the open-access journal *PLOS ONE* [20] using AntCorGen [21]. *PLOS ONE* publishes a diverse range of peer-reviewed papers across various scientific disciplines. AntCorGen is a free software tool designed to automatically generate corpora from online sources, thereby supporting a wide range of applications in linguistic research (e.g., [22,23]).

Figure 1 illustrates the data collection procedure using AntCorGen.

Samples from five disciplines were collected and analyzed: biochemistry (BIO), civil engineering (CIVIL), computer and information sciences (CIS), electronics engineering (ELECT), and mechanical engineering (MECH). These five disciplines were selected because they represent key areas in science and technology research, ensuring a balanced sample covering both the life sciences and applied sciences, as each encompasses life sciences and biological aspects, applied sciences, information technology and computational sciences, electrical and electronic systems, mechanics, thermodynamics, and materials science. This diversity allows a comprehensive analysis of how linguistic features manifest across disciplines with differing methodologies, terminologies, and research goals.

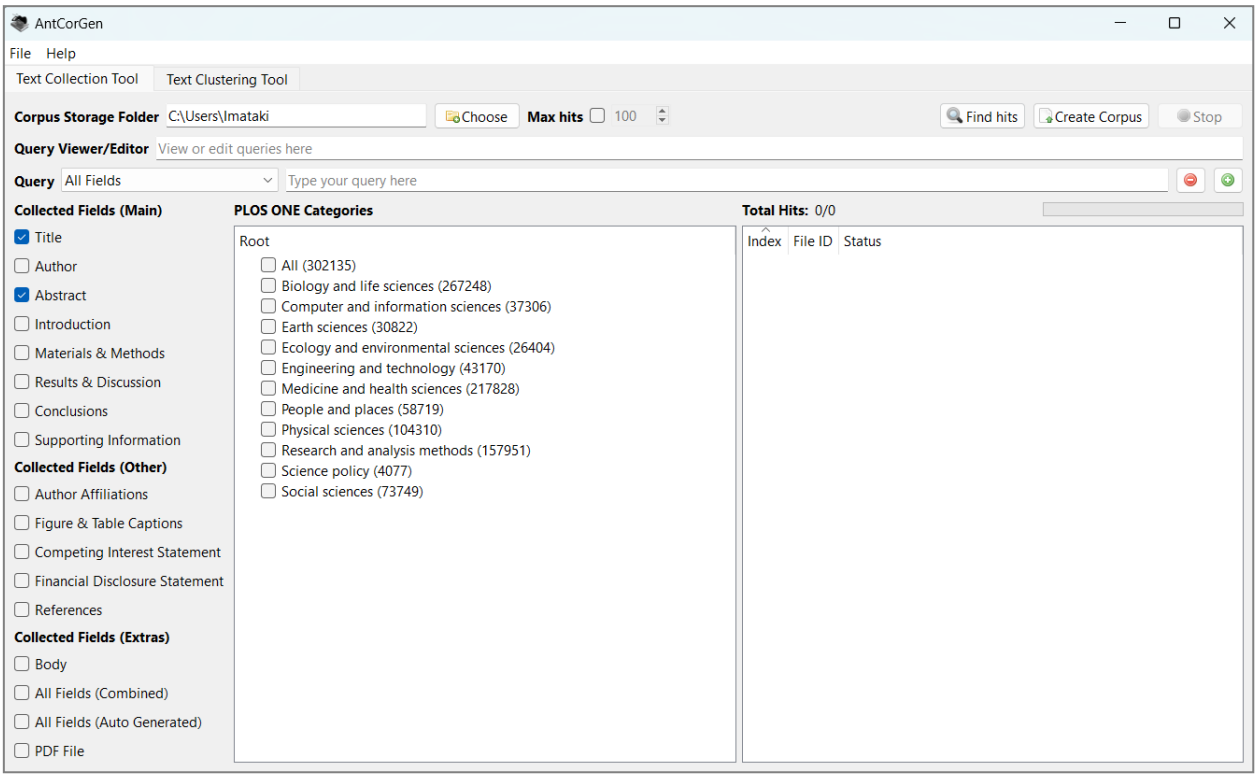


Figure 1. Data collection using AntCorGen

Table 1 summarizes of the created corpora. In this study, 1,000 abstracts from each of the five disciplines were randomly selected, and a corresponding corpus was created for each discipline. The number of words in each corpus ranges from 277,342 to 534,994.

Table 1. Summary of the corpora

Discipline	Number of texts	Words
BIO	1,000	534,994
CIVIL	1,000	291,583
CIS	1,000	281,716
ELECT	1,000	331,881
MECH	1,000	277,342
Total	**5,000**	**1,717,516**

2.2. Methods

2.2.1. The 67 Linguistic Features from Biber (1988)

Biber's [1] multidimensional analysis framework was originally proposed to analyze the differences between written and spoken English based on 67 linguistic features (See Appendix). This framework quantitatively analyzes language variation across different contexts by identifying the co-occurring linguistic features that define distinct dimensions of variation. Through factor analysis, Biber [1] identifies these dimensions, such as "informational vs. involved" discourse, which reveal how texts vary systematically based on their linguistic features. The framework is corpus-based and uses large text collections to compare language use across genres and registers. It has been used to analyze sub-genres of written language such as academic writing and fiction [24]. Additionally, it has been applied to analyze the differences in the linguistic characteristics of medical papers between chapters in British and American English [25], and to analyze texts that belong to Internet registers [26]. Therefore, it is considered applicable to feature analysis within the domain of scientific and technological papers.

A Multidimensional Analysis Tagger (MAT [27]) was used to calculate the frequency of each linguistic item. AntConc [28] was used to examine how each linguistic item appeared in context.

2.2.2. Statistical Processing

This study primarily applied correlation analysis and hierarchical cluster analysis as analytical methods whose purpose is to investigate the similarities and differences among the five academic disciplines. Both statistical analyses were conducted using frequency data obtained from the MAT.

Correlation analysis is a powerful statistical method used to examine the strength and direction of relationships between two or more variables and is commonly applied in linguistic research [29,30]. Correlation coefficients, ranging from -1 to +1, indicate the type of relationship between variables: values close to +1 suggest a positive correlation (both variables increase together), while those near -1 indicate a negative correlation (one variable increases as the other decreases). A value near 0 reflects little to no meaningful relationship between variables. Although correlation analysis typically examines two variables simultaneously, multiple variables can be analyzed simultaneously in a grid format. Several types of correlation coefficients exist, and the Pearson product-moment correlation coefficient is commonly used when analyzing frequency data.

Hierarchical cluster analysis (or hierarchical clustering) is an algorithm used to group similar objects into clusters based on shared attributes. This method organizes objects into a multilevel hierarchy, in which clusters at one level are progressively merged into larger clusters at subsequent levels. The result is a set of distinct clusters, in which the objects within each cluster exhibit high similarity, while the differences between clusters are maximized. This technique has been applied in various fields including linguistics [31]. While there are various options for performing cluster analysis, this study utilized standardized Euclidean distance and the complete linkage method. The distance and linkage methods were the default settings in R that we used for statistical analyses.

3. Results

Table 2 presents the correlation matrix. Each cell in the matrix displays the correlation between two variables. For example, the cell in which biochemistry and civil engineering intersect shows the strength and direction of the linear relationship between these two disciplines. Because all coefficients are close to 1 and similar in value, it can be interpreted that the five disciplines are highly similar. A coefficient close to one indicates a near-perfect positive correlation, suggesting that linguistic features in abstracts from these disciplines have a strong and direct relationship.

Table 2. Correlation matrix

Category	BIO	CIVIL	CIS	ELECT	MECH
BIO	1.0000	0.9995	0.9998	0.9995	0.9997
CIVIL	0.9995	1.0000	0.9997	0.9997	0.9996
CIS	0.9998	0.9997	1.0000	0.9998	0.9998
ELECT	0.9995	0.9997	0.9998	1.0000	0.9999
MECH	0.9997	0.9996	0.9998	0.9999	1.0000

From Table 2 overviewing the entire data set, there appear to be no substantial differences among the five disciplines. This suggests that abstracts from the five disciplines might share similar linguistic and structural features, such as common vocabulary, grammar, or

discourse patterns, possibly because of standardized academic writing conventions or interdisciplinary overlaps. Despite differences in content and subject matter, the way research is introduced, summarized, and discussed in abstracts across the five disciplines may follow a universal pattern of scientific communication. This could be because academic writing, especially research abstracts, tends to follow a formal structure (e.g., problem statements, methodology, results, and conclusion), contributing to similarity. Additionally, there may be a convergence in the research methods or analytical approaches used in these disciplines. If these disciplines rely on similar quantitative methods, technologies, or theoretical frameworks (e.g., computational modeling and statistical analysis), the language used in the abstracts could become more similar. These five disciplines may also share a substantial overlap or be influenced by interdisciplinary research. For instance, disciplines such as CIS and ELECT are closely related, and even BIO and CIVIL might use similar computational models or techniques, leading to similarities in how research is presented, even when the subject matter differs.

While the correlation matrix showed strong similarities across disciplines, cluster analysis was conducted to further investigate how these disciplines grouped together based on their linguistic characteristics. As a result, it was shown that there are disciplines that are relatively "close" or "distant" among the five. Figure 2 shows the results of the hierarchical cluster analysis.

Based on the frequency of the 63 out of 67 linguistic items analyzed in this study, the disciplines can be broadly categorized into two groups: BIO and CIS, and a second group consisting of MECH, CIVIL, and ELECT (four linguistic features were excluded from consideration because they were not observed in the corpora). In the latter group, civil engineering and electronics engineering are relatively similar.

The data shows that biochemistry has a higher frequency of many items than other disciplines, such as passive voice using *by* (BYPA), conjunctions (CONJ), demonstratives (DEMO), and attributive adjectives (JJ).

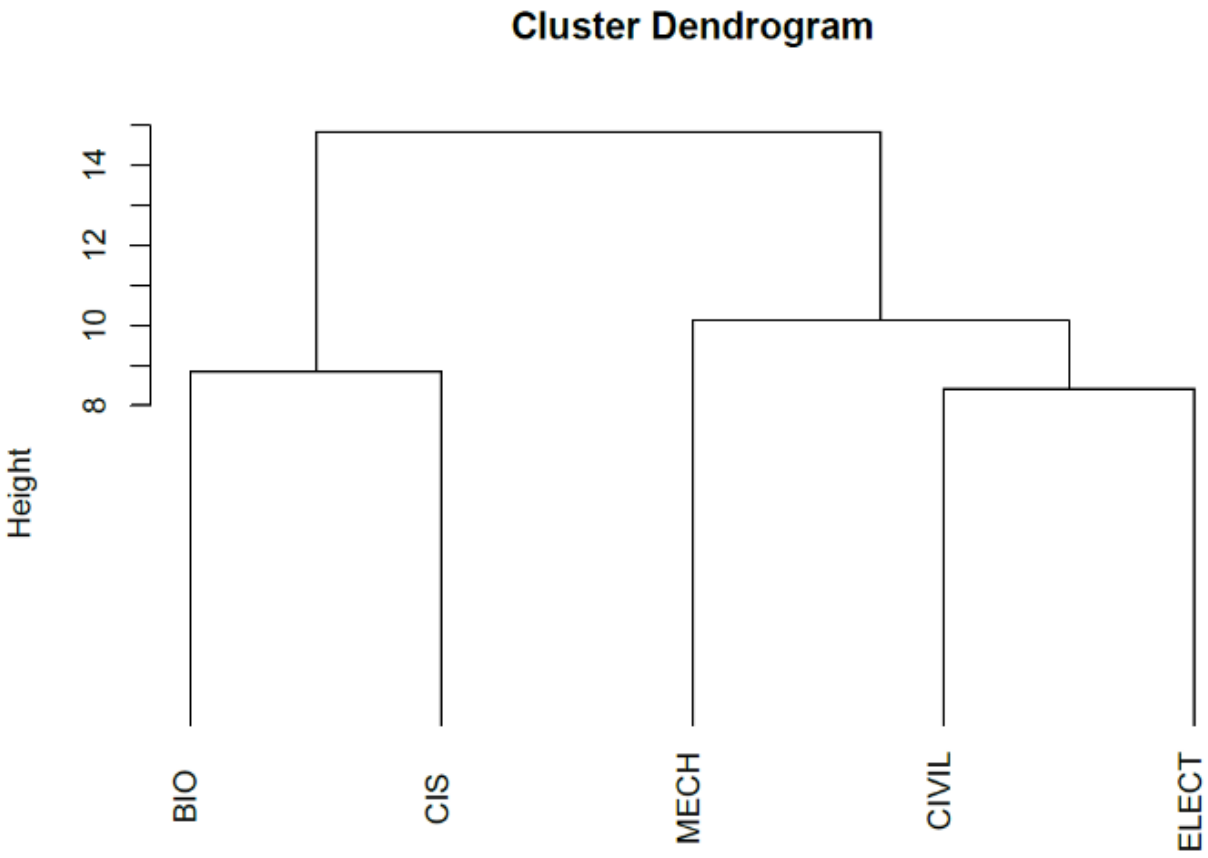


Figure 2. Hierarchical cluster analysis

Looking more closely at the passive voice using *by* in biochemistry, the phrase *is/are (also) regulated by* appears most frequently, with 45 out of 1,212 instances. Other frequently occurring phrases include *is/are caused by*, *is/are mediated by*, *is/are (also) characterized by*, and *is/are controlled by*, indicating that passive voice is often used to describe mechanisms and characteristics.

Regarding conjunctions in BIO, out of 3,028 instances, adversative *however* appears 697 times (23.0%), additive *furthermore* appears 256 times (8.5%), and causal *therefore* appears 253 times (8.4%). These three conjunctive adverbs accounted for 39.8% of all conjunctions used in BIO. Although there are no particular features in the proportion of individual vocabulary items or phrases, BIO abstracts are characterized by a wide variety of expressions.

For attributive adjectives, out of 64,325 instances in biochemistry, *binding* appeared most frequently, with 934 (1.5%) of the attributive adjectives used in BIO abstracts (e.g., *DNA binding motifs*, *DNA binding proteins*). This was followed by *specific* with 869 instances (1.4%, e.g., *the specific DNA target sequence*) and *other* with 802 instances (1.2%, e.g., *other proteins*, *other species*, *other types*). This may reflect the tendency in biochemistry to use large-noun phrases consisting of multiple words to describe substances and phenomena.

Additionally, BIO and CIS had higher frequencies of amplifiers (AMP). In BIO, of the 909 instances, *highly* appears 353 times (38.8%), *very* appears 155 times (17.1%), *strongly* appears 122 times (13.4%), *fully* appears 83 times (9.1%), and *completely* appears 52 times (5.7%). These five adverbs account for 84.2% of the amplifiers used in BIO.

In CIS, out of 443 instances, *highly* appears 153 times (34.5%), *very* appears 103 times (23.3%), *strongly* appears 60 times (13.5%), *fully* appears 39 times (8.8%), and *completely* appears 23 times (5.2%). These five adverbs accounted for 85.3% of the amplifiers used in CIS.

BIO and CIS also have a higher frequency of private verbs (PRIV), such as *show*, *found*, *known*, *demonstrate*, and first-person pronouns (FPP1), such as *we*, *our*, or *us*, compared to other disciplines. In contrast, CIVIL shows a higher frequency of other nouns (NN) as subjects (e.g., *The goal/aim of this study is to*, *These data are analyzed*, or *These results show*).

Both BIO and MECH exhibited a higher frequency of the pronoun *it* (PIT).

Moreover, BIO has a longer average word length (AWL), likely because of the frequent use of proper nouns and technical terms compared to other disciplines. By contrast, MECH has a lower type-token ratio (TTR), meaning that abstracts have less vocabulary variety.

BIO also had fewer past tense verbs, with a high frequency of present tense verbs (VPRT), totaling 25,415 instances. Among these, the verb *be* appears 3,581 times, including in passive constructions. This is followed by *have/has* with 2,423 instances, which includes the present perfect tense.

The prominence of passive constructions in BIO reflects the need to describe processes and experiments objectively, distancing researchers from action. However, the frequent use of conjunctions like *however* and *therefore* may indicate complex argument and causal relationships that are central to the field's research methodology.

ELECT shows a higher frequency of specific negation expressions, such as synthetic negation (SYNC), such that *no significant differences were found*.

MECH exhibited a higher frequency of emphatics (EMPH) and adverbial phrases indicating location (PLACE) with a lower frequency of attributive adjectives (JJ). It involves a high frequency of *there*-constructions (EX).

Additionally, CIS shows the highest frequency of the verb *be* used as the main verb (BEMA) among the five disciplines (e.g., *although well established, this method is time-consuming and error-prone*).

4. Discussion

By analyzing the frequency of 63 linguistic features, this study identified key patterns in how language is used across the five disciplines. These patterns reveal both the commonalities and distinctive features that characterize specific communication practices in each discipline. This section discusses these findings and their implications for scientific communication and education.

The five disciplines can be categorized into two broad groups: biochemistry and computer and information sciences, and a second group consisting of mechanical engineering, civil engineering, and electronics engineering. The close relationship between civil engineering and electronics engineering suggests that findings and methods can be more easily transferred between these two disciplines, possibly because of shared methodologies and a focus on infrastructure and systems. The clustering of biochemistry with computer and information sciences, for instance, might be attributed to the reliance on precise technical vocabulary in both fields, where specificity is key to describing processes and results. However, the grouping of other engineering disciplines may reflect shared methodologies and reporting structures, particularly focusing on system-based research.

In biochemistry, the frequent use of the passive voice, particularly with *by*, and conjunctions like *however* and *therefore* reflects the need to describe processes objectively and demonstrate logical relationships between ideas. According to Leong [32], approximately 30% of all clauses in scientific research articles are written in the passive voice. Our findings suggest that biochemistry papers may use passive constructions more frequently than papers in other disciplines. The high frequency of large noun phrases highlights the importance of specificity and clarity in naming substances and processes, which are important for accurately communicating complex biochemical concepts.

The frequent use of amplifiers and private verbs in

biochemistry and computer and information sciences may reflect the need to emphasize the significance of findings in these disciplines. As Quirk et al. [33] mentioned, private verbs are used to express intellectual states such as beliefs and intellectual acts such as discovery. Researchers can use these features to emphasize the importance of their results, particularly when presenting novel or breakthroughs. Additionally, Xiao & Tao [34] noted that amplifiers were more common in speech than in writing. In this context, the writing style for biochemistry and computer and information sciences can be characterized as being closer to spoken language than those in the other three disciplines.

These insights can inform the development of better-targeted educational strategies for ESP courses. For example, teaching passive constructions and technical specificity is essential for biochemistry students, for whom objectivity and clarity are regarded as highly important. Meanwhile, in computer and information sciences discipline, students should be guided to use first-person pronouns and private verbs appropriately, as these disciplines often require a more direct involvement of the researcher in the narrative.

However, these analyses were based on a specific set of abstracts, and may not capture the full range of language use across disciplines. Future research could expand this work by incorporating additional rhetorical features and examining how these features impact reader comprehension and engagement.

In summary, this study provides insights into the linguistic features of abstracts across five scientific and technical research disciplines. By identifying the specific patterns and trends in each discipline, we offer a foundation for more effective communication strategies tailored to the unique needs of each discipline [35–37]. These insights can also inform the development of writing instruction programs that address the universal and discipline-specific needs of students and researchers.

5. Conclusions

This study provides valuable insights into the linguistic features of abstracts across five scientific and technical disciplines. By analyzing 63 linguistic features, this study identified both commonalities and differences between disciplines, revealing that fields such as biochemistry and computer and information sciences share certain linguistic patterns, whereas engineering disciplines tend to cluster together. These findings suggest that, while some linguistic features are universal in scientific writing, others are shaped by the specific demands and methodologies of each discipline.

The results have important implications for ESP instruction, particularly in higher education. By understanding the linguistic conventions of different fields, educators can tailor their teaching to address the universal and discipline-specific needs of students. For example, the frequent use of passive voices in biochemistry and the emphasis on first-person pronouns in computer and information sciences highlight the need for specialized guidance in writing in these fields.

Despite its contributions, this study had several limitations. The analysis was based on a specific set of linguistic features, and future research could expand this work by exploring additional rhetorical elements, such as argumentation patterns or citation practices. Furthermore, examining how these features affect reader comprehension could provide deeper insights into effective scientific communication.

Overall, this study lays the foundation for a more effective ESP instruction and a cross-disciplinary understanding of scientific writing. By exploring the intersection between language and disciplinary norms, educators and researchers can contribute to the ongoing improvement of scientific communication and academic writing instruction.

Appendix: The 67 Linguistic Features from Biber (1988)

A. Tense and aspect markers
1. Past tense (VBD), 2. Perfect aspect (PEAS), 3. Present tense (VPRT)
B. Place and time adverbials
4. Place adverbials (PLACE), 5. Time adverbials (TIME)
C. Pronouns and pro-verbs
6. First person pronouns (FPP1), 7. Second person pronouns (SPP2), 8. Third person personal pronouns (excluding *it*) (TPP3), 9. Pronoun *it* (PIT), 10. Demonstrative pronouns (DEMP), 11. Indefinite pronouns (INPR), 12. Pro-verb *do* (PROD)
D. Questions
13. Direct WH-questions (WHQU)
E. Nominal forms
14. Nominalizations (ending in *-tion, -ment, -ness, -ity*) (NOMZ), 15. Gerunds (GER), 16. Total other nouns (NN)
F. Passives
17. Agentless passives (PASS), 18. *by*-passives (BYPA)
G. Stative forms
19. *be* as main verb (BEMA), 20. Existential *there* (EX)
H. Subordination features
21. *that* verb complements (THVC), 22. *that* adjective complements (THAC), 23. WH clauses (WHCL), 24. Infinitives (*to*-clause) (TO), 25. Present participial clauses (PRESP), 26. Past participial clauses (PASTP), 27. Past participial WHIZ deletion relatives (WZPAST), 28. Present participial WHIZ deletion relatives (WZPRES), 29. *that* relative clauses on subject position (TSUB), 30. *that* relative clauses on object position (TOBJ), 31. WH relatives on subject position (WHSUB), 32. WH relatives on object position (WHOBJ), 33. Pied-piping relative clauses (PIRE), 34. Sentence relatives (SERE), 35. Causative adverbial subordinators (*because*) (CAUS), 36. Concessive adverbial subordinators (*although, though*) (CONC), 37. Conditional adverbial subordinators (*if, unless*) (COND), 38. Other adverbial subordinators (OSUB)
I. Prepositional phrases, adjectives, and adverbs
39. Total prepositional phrases (PIN), 40. Attributive adjectives (JJ), 41. Predicative adjectives (PRED), 42. Total adverbs (RB)
J. Lexical specificity
43. Type/token ratio (TTR), 44. Mean word length (AWL)
K. Lexical classes
45. Conjuncts (CONJ), 46. Downtoners (DWNT), 47. Hedges (HDG), 48. Amplifiers (AMP), 49. Emphatics (EMPH), 50. Discourse particles (DPAR), 51. Demonstratives (DEMO)
L. Modals
52. Possibility modals (POMD), 53. Necessity modals (NEMD), 54. Predictive modals (PRMD)
M. Specialized verb classes
55. Public verbs (PUBV), 56. Private verbs (PRIV), 57. Suasive verbs (SUAV), 58. *seem* and *appear* (SMP)
N. Reduced forms and dispreferred structures
59. Contractions (CONT), 60. Subordinator *that* deletion (THATD), 61. Stranded prepositions (STPR), 62. Split infinitives (SPIN), 63. Split auxiliaries (SPAU)
O. Coordination
64. Phrasal coordination (PHC), 65. Independent clause coordination (ANDC)
P. Negation
66. Syntactic negation (SYNE), 67. Analytic negation (XX0)

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